AI-Based Patient Activity Monitoring System for Nursing Support

MSc Dissertation - Masters Project

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Presentation Context

What this project is

"A computer-vision pipeline that watches a patient in a hospital ward, understands what they are doing, and tells nursing staff — in plain language — when attention is needed."

This is a decision-support prototype — it is not a replacement for clinical judgement.

Throughout this presentation, I will explain the problem, the system design, the methodology, the results, and the future improvements of the project.

Project Overview

The Problem

- Hospital nurses are responsible for continuous patient observation but cannot physically watch every patient in a ward simultaneously.
- Patients who are at risk — from falls, pressure injuries, or unexpected mobility — may go unnoticed for critical minutes.
- Existing monitoring solutions either rely on invasive wearable sensors or expensive bespoke hardware.
- There is a clear gap for a camera-based, non-contact system that can interpret what a patient is actually doing.

Why It Matters

- Falls are the most common patient safety incident in UK hospitals, costing the NHS an estimated £2.3 billion annually.
- Activity monitoring can support repositioning schedules, early mobilisation tracking, and fall risk escalation.
- A computer-vision approach requires no patient contact, works with existing CCTV infrastructure, and scales to multiple beds.
- This project delivers a working prototype that demonstrates feasibility at low cost and with publicly available data.

Aims and Objectives

Aim

To design, build, and evaluate a multi-layer computer-vision pipeline that can automatically recognize patient activities in a hospital ward setting from camera footage, and communicate the results in a format suitable for nursing staff.

Objectives

- Acquire and analyse suitable annotated hospital-ward image datasets to understand class distribution and data quality.
- Train and evaluate a YOLOv8 object-detection model on patient activity classes (fall, move, lie, sit, stand), distinguishing them from context and equipment objects.
- Integrate a pose-estimation layer to extract skeletal keypoints and derive geometric cues — posture, face direction, and hand-to-face proximity.
- Apply spatial reasoning to understand the patient's position relative to the hospital bed.
- Implement a temporal state machine that aggregates frame-level evidence into clinically meaningful activity labels (sleeping, drinking, fall, restless, etc.).
- Conduct a systematic experiment suite comparing model sizes, confidence thresholds, and test-time augmentation to select the optimal configuration.
- Produce nurse-readable overlays and a structured JSON activity log as the deliverable output.

Background and Motivation

Clinical Context

- NHS Improvement reports ~250,000 inpatient falls per year in England alone.
- Manual patient observation rounds are typically 30–60 minutes apart — a long gap for high-risk patients.
- Pressure injuries caused by prolonged static posture cost the NHS approximately £1.4 billion annually.

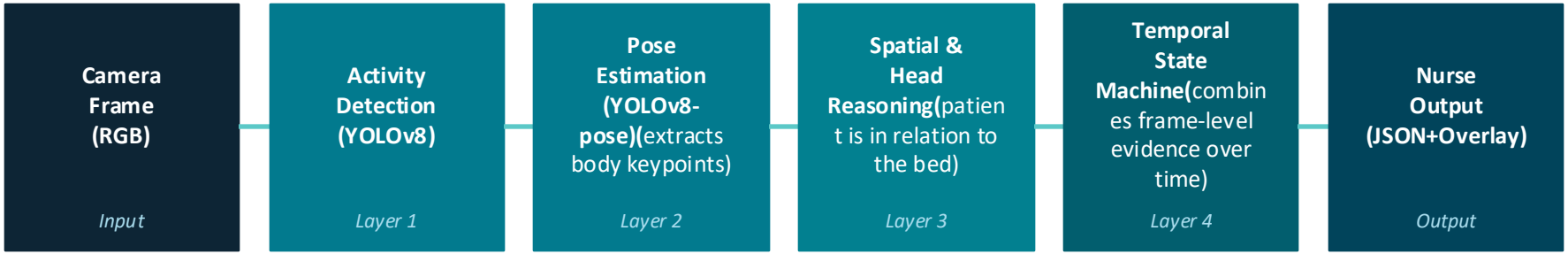
Prior Technical Work

- Wearable sensors (accelerometers, gyroscopes) provide reliable fall detection but require patient compliance and charging.
- Depth cameras (Kinect, lidar) work well but are costly and require specific installation.
- RGB-only vision approaches have advanced significantly with transformer-based detectors and pose models.

Gap This Project Fills

- Existing open-source pipelines focus on general action recognition, not specifically on nursing-relevant patient activities.
- Few systems combine activity detection, pose, and spatial reasoning into a single deployable pipeline.
- This project delivers that integrated solution using only a standard RGB camera and an open dataset.

Methodology and System Architecture



Research Design Rationale

- Experimental approach: trained and compared three YOLOv8 model sizes (nano/small/medium) across 10 configurations — varying confidence threshold and test-time augmentation — then selected the winning setup.
- Datasets: hospital_bed activity dataset (Roboflow, 2,004 images, 11 classes) for training; pretrained COCO YOLOv8-pose for pose estimation.
- Ethical, social, and legal considerations: patient privacy is addressed by keeping all images anonymised within a research dataset; no real patient data was collected. GDPR compliance assumed for any production deployment. Clinical validity must be confirmed by domain experts before use.
- Tools: Python 3.11, PyTorch 2.1, Ultralytics YOLOv8, OpenCV, scikit-learn, NVIDIA RTX A4000 GPU.

Datasets and Exploratory Analysis

Dataset Overview

Dataset	Classes	Train	Val	Test	Role in Project
hospital_bed (activity)	11	1,401	402	201	Main — trained from scratch
hospital (equipment)	25	5,753	1,009	521	EDA reference only
bed_elevation	4	51	13	0	Too small — kept for reference

Activity Class Distribution — Training Split (hospital_bed dataset)

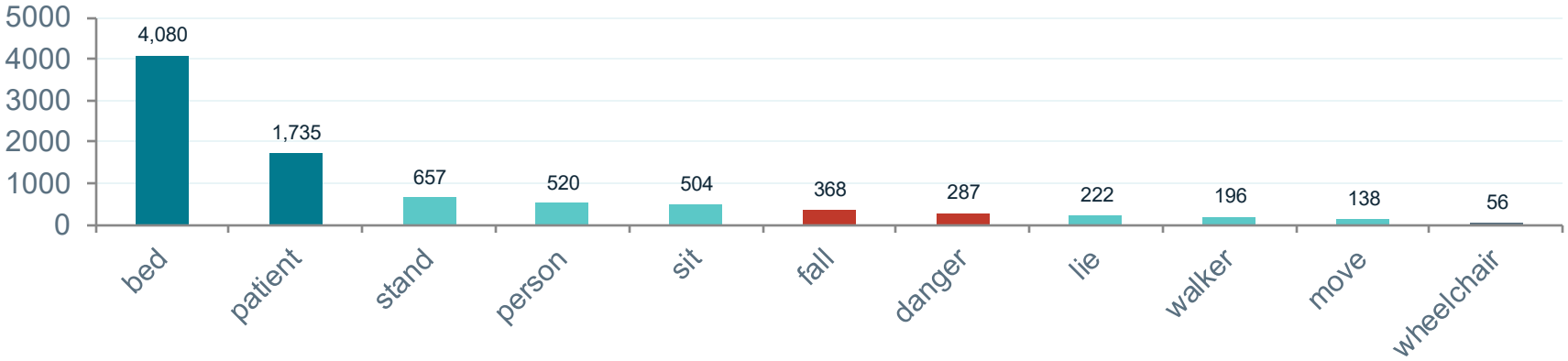


Figure 1. The dataset is imbalanced — bed and patient instances dominate, while fall, danger, walker, and wheelchair are rare. YOLO's built-in class weighting compensates for this during training.

Implementation Summary

Model Training

Three YOLOv8 variants (nano, small, medium) fine-tuned for 30 epochs at 640x640 on GPU. Training used the AdamW optimiser with automatic early stopping (patience=10). All training runs were seeded for reproducibility.

Pose Layer

A pretrained YOLOv8-pose model extracts 17 COCO skeleton keypoints per person. Keypoints with confidence below 0.4 are masked to avoid unreliable geometric reasoning.

Spatial Reasoning Module

Bounding-box geometry determines whether a patient is on the bed (IoU threshold 0.15) and which side of the bed they are on (centroid offset from the bed centre). Designed to fail gracefully when the bed is not visible.

Head Pose & Hand Proximity

Face direction (frontal, turned left/right, face-down) is derived from nose and ear keypoint visibility. Hand-to-face proximity acts as a drinking proxy when a cup or bottle is also detected.

Temporal State Machine

A sliding-window state machine (window=30s at 1 fps) aggregates frame-level facts into activity labels: fall_detected, sleeping, drinking, restless_in_bed, mobilising_with_aid. Fall detection fires on any single frame — all other states require sustained evidence.

Output & Logging

The MonitorPipeline class produces: (a) annotated video frames with YOLO boxes, skeleton overlay, and a nurse-readable banner; (b) a structured JSON activity log per frame with patient_state, fall_risk, bed_status, side_of_bed, face_direction, and nurse_action fields.

Detection Results — Validation Metrics

94.3%
MAP50(Mean Average Precision) (all classes)

77.3%
mAP50-95 (all classes)

91.8%
Mean Precision(how many of the model's detections were correct)

89.2%
Mean Recall(how many actual objects or activities the model successfully found.)

Per-Class AP50 (Validation Split)

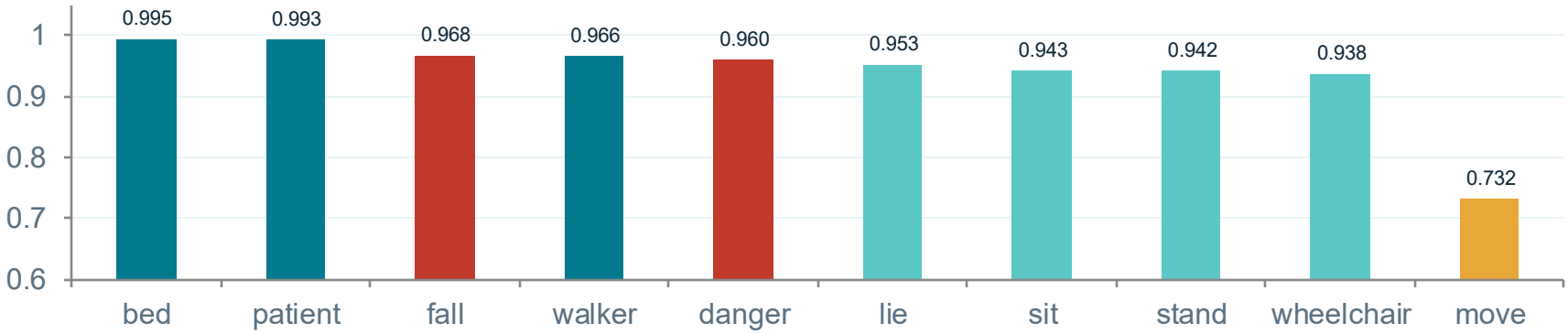
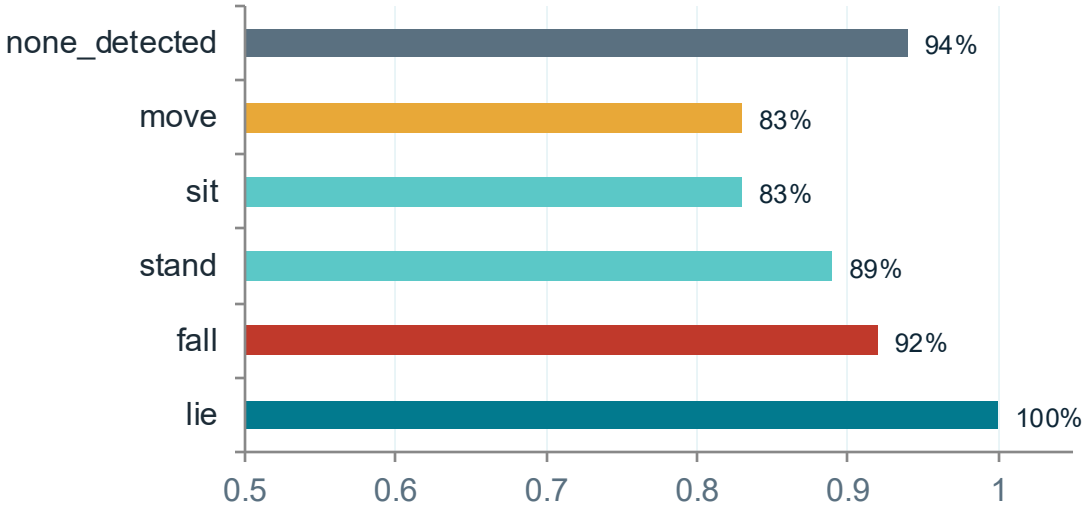


Figure 2. Per-class AP50 on the validation split. Critical safety classes (fall, danger) score above 0.96. The move class scores lower at 0.73 — expected, as motion is an action class with no clear visual signature.

Activity Recognition Results — Primary Metric



91.0%

Activity Recognition Accuracy
(val split, winning model)

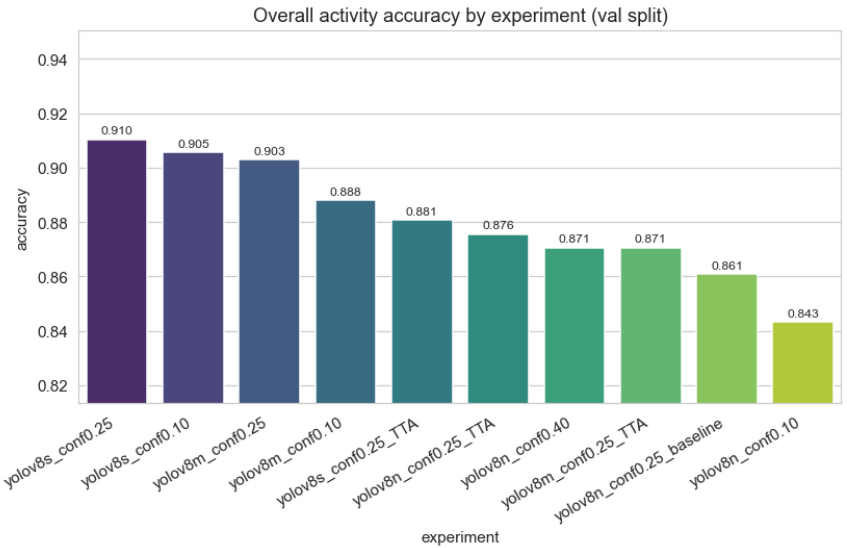
Key Findings

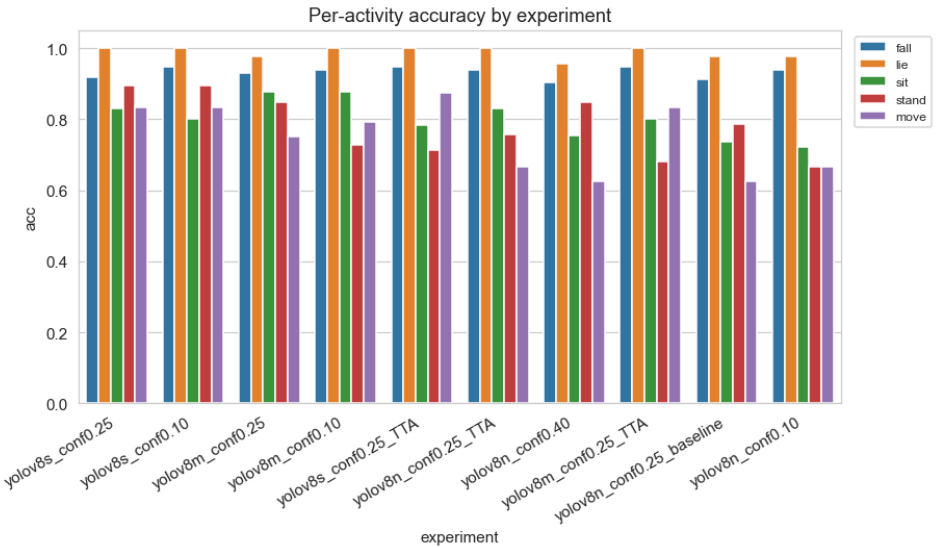
- Fall detection achieves 92% per-class accuracy — the most critical result for patient safety.
- Lie class is near-perfect at 100%; the lying posture is visually distinctive and common in the dataset.
- Move scores lowest at 83% — it is an action class without a clean single-frame visual signature.

Figure 3. Per-activity accuracy (val split, YOLOv8s conf=0.25). Validation split used — test split is biased toward fall/danger only. Danger flag accuracy: 96.5% separately.

Experiment Suite — Finding the Optimal Configuration

10 configurations compared: 3 model sizes x 3 confidence thresholds x Test-Time Augmentation (on/off)





Winner

- Winner: YOLOv8s at conf=0.25 (no TTA) — 91.04% activity accuracy, fall accuracy 92%.
- TTA produced a negative result: with small backbones, augmented predictions disagree more than they agree on borderline cases, reducing accuracy rather than improving it.
- Model size: YOLOv8m did not meaningfully outperform YOLOv8s, so the smaller model was chosen for better inference speed.

Discussion and Key Insights

- The **multi-layer architecture justifies itself:** YOLO alone confuses 'lie' with 'fall' in low-angle frames, but the pose layer's keypoint geometry provides a disambiguating second signal. Neither layer is sufficient on its own.
- The **move class is the clear weak point**. Motion is an inherently temporal concept — a single-frame detector cannot fully capture it. The state machine partially compensates but cannot recover lost per-frame accuracy.
- TTA(**test-time augmentation**) was expected to improve accuracy and did not. This was the result that most changed my approach: it led me to investigate why (small models + priority-voting logic amplifies disagreement) and informed the decision not to use TTA in the final pipeline.
- The **validation-vs-test split** choice revealed a dataset quality issue: the Roboflow test split is biased toward fall/danger. Using the validation split as the primary evaluation set was the correct methodological response and is documented as a limitation.
- The state machine's sleeping threshold (85% lying + low motion + 60% closed eyes) is necessarily heuristic. In a real deployment, these thresholds would need clinical calibration against ground-truth shift-change observations.
- Danger detection at 96.5% is surprisingly strong given that 'danger' describes a scene condition, not a patient posture — this suggests the visual signature of hazard scenes (e.g. lowered bed rails) is distinctive enough for YOLO to learn.

Value, Contribution, and Challenges

Project Contribution

- A fully integrated, end-to-end patient activity monitoring pipeline — not just a trained model.
- A systematic experiment report comparing 10 configurations, producing a reproducible best configuration.
- A nurse-friendly output format (structured JSON + annotated overlay) that bridges the gap between AI output and clinical practice.
- A demonstration that a usable system can be built from a single open dataset, open-source models, and a single GPU.
- Stakeholder impact: a deployed version could support nurse staffing decisions, reduce fall-related incidents, and provide an audit trail for repositioning care.

Challenges Overcome

- Dataset imbalance was handled using YOLO's built-in loss balancing instead of manual resampling.
- The Roboflow test split was biased toward fall/danger images, so validation results were used as the main metric.
- TTA reduced performance, so experiments were rerun to confirm the result.
- Sliding-window thresholds needed tuning to balance false alarms and missed events.
- Occluded pose detections were managed by masking low-confidence keypoints below 0.4.

Reflection — What I Learned and Would Do Differently

Key Learning

- Building a multi-layer pipeline revealed that integration complexity often exceeds the difficulty of any individual component. The state machine, in particular, required more iteration than the training step.
- The experiment suite demonstrated that intuitive assumptions (TTA will help; bigger model is always better) need to be tested rather than assumed — a lesson in disciplined empirical research.
- Dataset quality assessment is as important as model architecture. Discovering the test-split bias mid-project required reworking the evaluation methodology — which ultimately made the results more credible.

Three Things I Would Do Differently

1 Collect a video dataset from the start

The still-image dataset forces the temporal state machine to treat each image as independent. Real ward video would enable proper temporal tracking and substantially improve the move class performance.

2 Involve clinical staff earlier

The state machine thresholds for sleeping and restlessness were set heuristically. A brief consultation with a nursing team at the design stage would have grounded those values in observed clinical practice.

3 Implement a calibration mode

The bed-relative spatial reasoning uses camera-frame coordinates. A two-point bed calibration step (head-end and foot-end annotation at installation) would make left/right readings meaningful in bed coordinates rather than camera coordinates.

Future Work

- Video dataset collection: partnering with a hospital trust to collect anonymised ward video under an ethics protocol. Even a modest 50-hour annotated dataset would dramatically improve the move class and enable proper temporal training.
- Transformer-based detection: evaluate YOLOv9 and RT-DETR backbones on the same activity dataset to assess whether attention-based architectures improve rare-class recall (fall, move) without sacrificing inference speed.
- Real-time streaming pipeline: extend the MonitorPipeline to process RTSP camera streams, adding a Kafka or MQTT message queue to push nurse alerts to a ward dashboard in near real time.
- Multi-camera fusion: in a ward with multiple bed angles, fusing detections from two or more camera viewpoints would resolve the occlusion failures seen in the current single-camera design.
- Clinical validation study: a formal sensitivity/specificity study with ground-truth nursing observation data, ideally in a simulated ward environment, to establish the system's real-world reliability before any deployment.
- On-device inference: quantise the winning model (YOLOv8s) to INT8 to enable deployment on edge hardware (NVIDIA Jetson Orin) co-located with the camera, removing the need for a central GPU server.

Conclusion

- The project built an end-to-end AI pipeline for recognising hospital patient activities from RGB camera images. It achieved **91% activity recognition accuracy and 92% fall detection accuracy** on validation data.
- The **five-layer architecture** was important because activity detection, pose estimation, spatial reasoning, head-pose inference, and temporal logic worked better together than alone.
- Experiments across 10 configurations supported the final choice of **YOLOv8s with a 0.25 confidence threshold and no TTA.**
- JSON logs and annotated overlays make the system more usable for clinical documentation and review.
- The prototype is a strong proof of concept and could support future ward-level deployment after video testing, clinical calibration, and real-time streaming integration.

Thank You

Headline results — a reminder:
Activity recognition accuracy: 91.0% (YOLOv8s, conf=0.25, val split)
Fall detection accuracy: 92.0%
Danger flag accuracy: 96.5%
Detection mAP50 / mAP50-95: 94.3% / 77.3%

Questions?

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